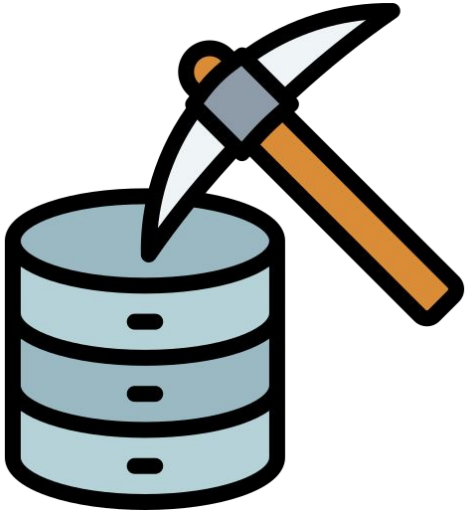
Data Mining in uber pickup: identifying spatial patterns, temporal trends and predictive insights



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Project Link:
colab.research.google.com/drive/1qP5UKOGPOE3PLKUCvkfQ1scLE3k46L0L#scrollTo=XQf3HGyeZHBQ

INTRODUCTION

On-demand private transportation (e.g., Uber) is a key part of urban mobility, especially in NYC.
Large fleets impact **traffic patterns** and **congestion**; understanding pickups helps optimize operations.

Project aims:

1. Identify **spatial demand hotspots**.
2. Analyze **temporal trends** (hourly, daily, monthly).
3. Build a **predictive model** to support fleet allocation

METHODS AND ALGORITHMS

Tools & Pre-processing

- Pandas & NumPy → load/merge data, convert Date/Time,
  → create temporal features (hour, day, month)
- Scikit-learn → clustering & regression

Spatial Analysis – K-Means

- Identify **10 pickup zones** using latitude & longitude
- Centroids = central points of each cluster

Visualization & Density – Folium

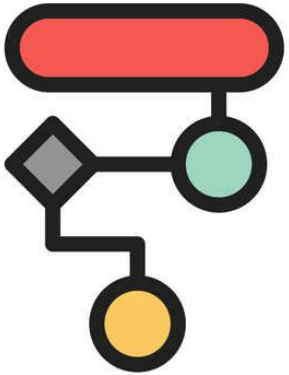
- Interactive maps of **hotspots**
- Markers & popups to highlight high-demand areas

Flow Analysis – NetworkX

- Model transitions between clusters as **undirected graphs**
- Nodes = clusters, edges = pickup flows, weights = frequency

Predictive Model – Random Forest

- Forecast **hourly pickups** based on temporal features
- Robust, handles non-linear trend



DATA DESCRIPTION

Source: Kaggle (2014 Uber pickups)
Focus: May–September 2014
Monthly Files:

- uber-raw-data-may14.csv … uber-raw-data-sep14.csv

Columns:

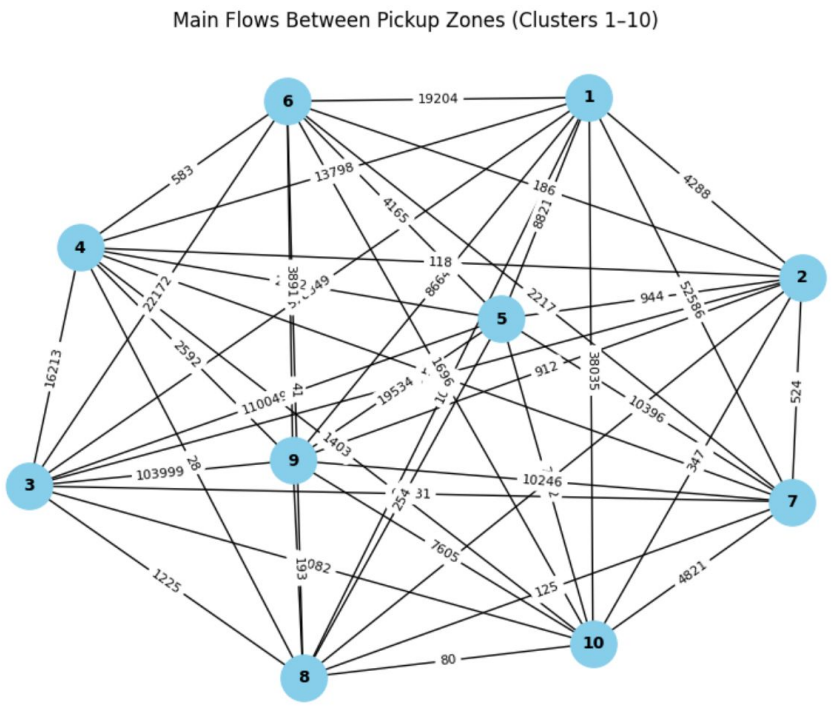
- Date/Time → timestamp of pickup
- Lat / Lon → pickup location
- Base → operational base ID

Dataset Size & Use:

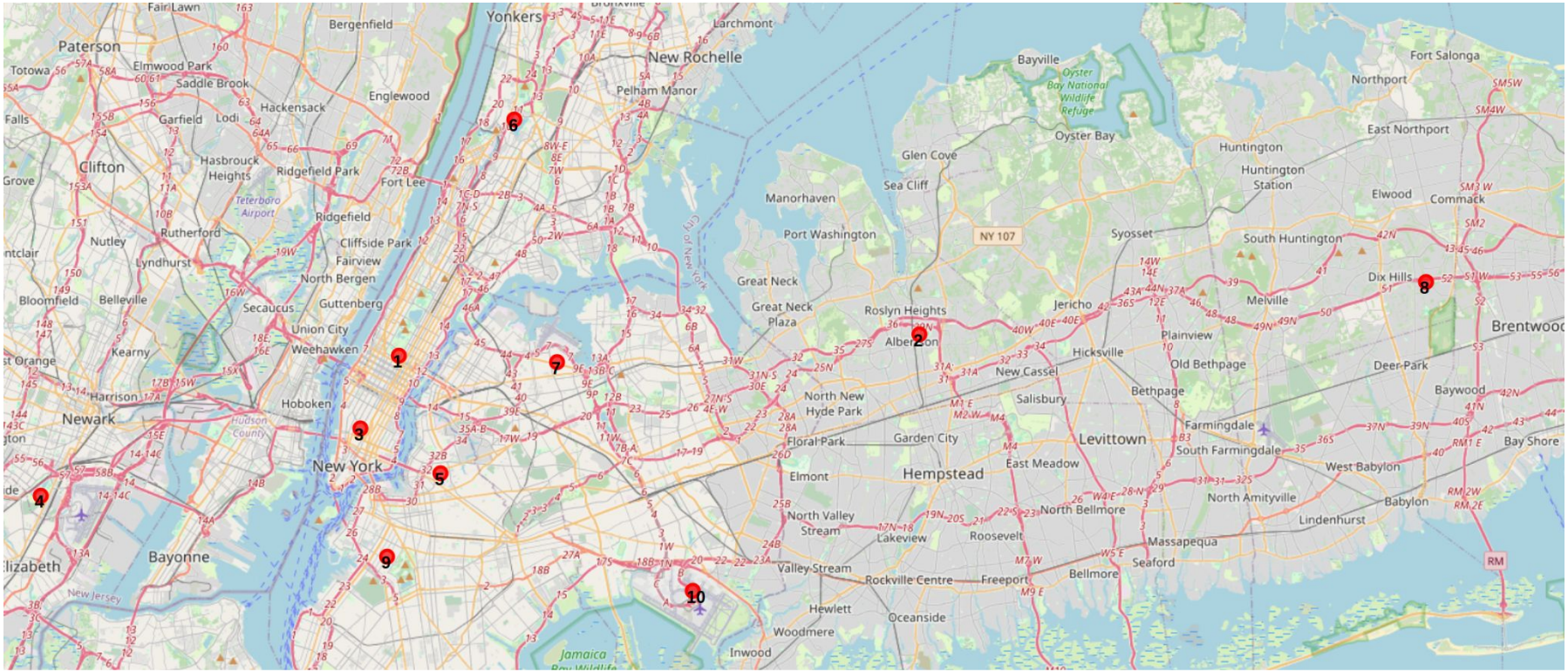
- Hundreds of thousands of pickups per month
- Suitable for spatial-temporal analysis, clustering, and predictive modeling

DATA EXPLORATION

The figures show both a graph-based and a map-based representation of the pickup distribution, allowing a clear visualization of the areas with the highest activity within the city.

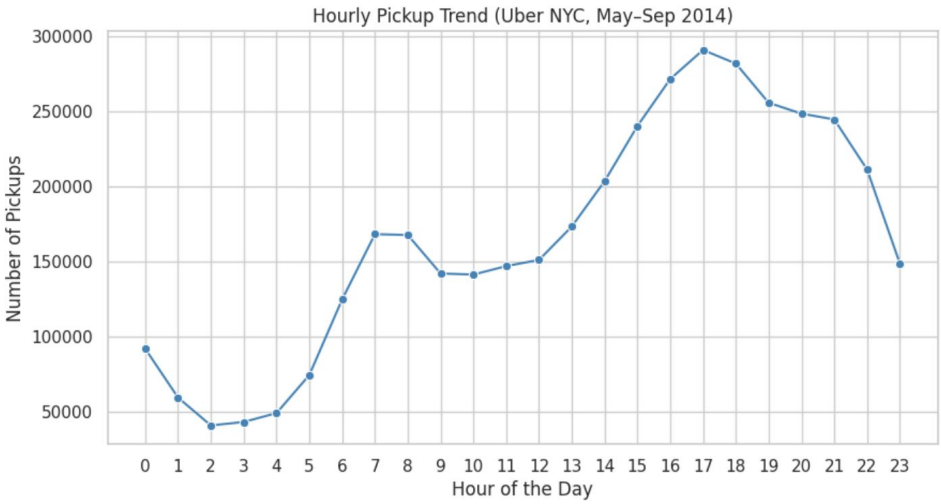


DATA EXPLORATION



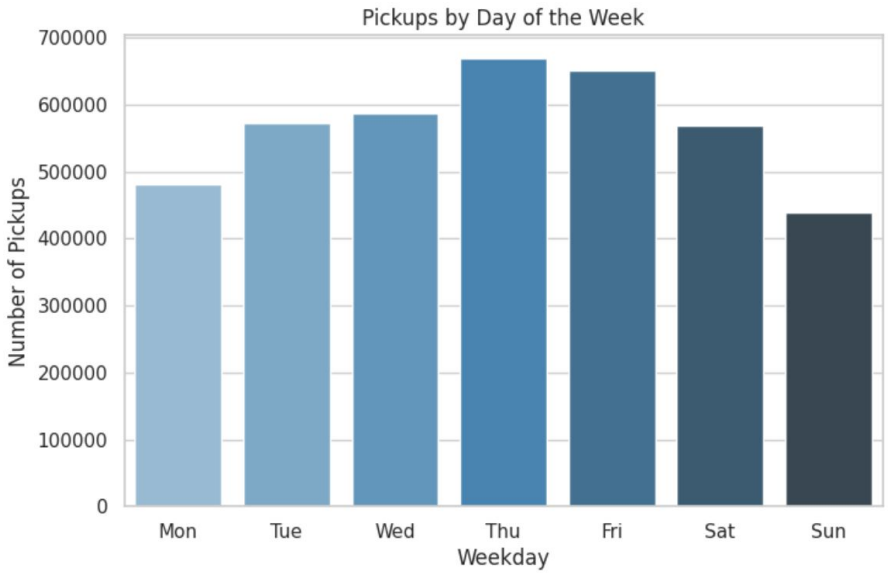
DATA EXPLORATION

Uber activity shows a clear daily trend, influenced by commuting flows and evening activities.



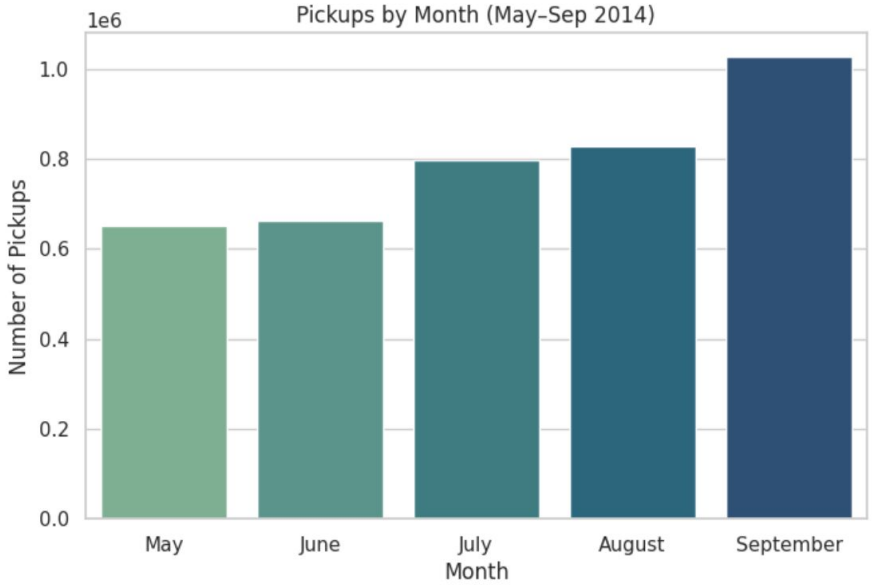
DATA EXPLORATION

Weekly analysis shows a clear distinction between weekdays and weekends:



DATA EXPLORATION

The monthly analysis highlights the number of pickups recorded in each month between May and September 2014, clearly showing that the months with the highest number of rides are August and September.



MODELING AND RESULTS

This figure shows the scatter plot of the data points, where the X-axis represents the actual number of pickups (Actual Pickups) and the Y-axis represents the predicted number (Predicted Pickups).

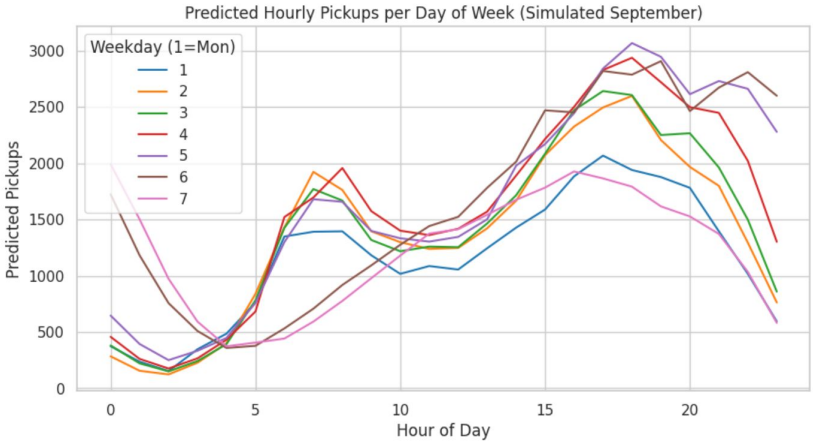
The goal is for the points to cluster as closely as possible along the y = x diagonal, which represents a perfect match between prediction and reality.

The close alignment of the points to the diagonal, especially for values up to 2000 pickups, indicates the high predictive accuracy of the model.

MODELING AND RESULTS

This picture presents a simulation of how hourly demand (Predicted Pickups) varies depending on the hour of the day (Hour of Day) and the day of the week (Weekday, where 1=Monday and 7=Sunday).

This graph allows us to identify clear commuting peaks (morning and evening) on weekdays and higher, more prolonged nighttime and leisure demand patterns typical of weekends (lines 6 and 7)



MODELING AND RESULTS

Model Performance – Random Forest

- **Mean Squared Error (MSE):** 60,202 → indicates average prediction deviation (large scale of pickups)

- **R^2:** 0.854 → model explains **85% of variability** in hourly pickups → **high predictive accuracy**

- **Interpretation:**
  - Predictions closely match actual demand
  - Features like **hour, weekday, and month** effectively capture demand patterns

CONCLUSION

Key Contributions:

- Spatial clustering → **10 demand hotspots**
- Graph analysis → **mobility flows**
- Machine learning → **Random Forest model** (R^2 = 0.854) predicting hourly pickup demand
- Identification of **peak times**: evenings & Thursday

Operational Implications:

- **Fleet allocation:** optimize vehicles & reduce waiting times
- **Urban planning:** Interpret demand patterns to anticipate congestion in key zones & time slots

Future Work:

- Include **external variables**: weather, events, holidays, pricing
- **Fine-grained analysis**: micro-hotspots & intra-zone flows